

VIRIKA OLIVIA SOANS

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PROFESSIONAL SUMMARY

Results-driven technology enthusiast with expertise in Python, SQL, data analytics, and software development. Skilled in designing and implementing efficient solutions, analyzing complex datasets, and developing applications that drive informed decision-making. Experienced in data processing, database management, and software engineering principles, with a strong foundation in problem-solving, algorithmic thinking, and delivering quality technical solutions.

CORE COMPETENCIES

Analytics & ML: Predictive Modelling, Statistical Analysis, Data Visualization, Deep Learning, Explainable AI

Tools & Platforms: Python, SQL (MySQL), Power BI, Jupyter, GitHub, Cloud Platforms

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Gradio, Flask

Other: HTML, CSS, Frontend Development

PROJECTS & RESEARCH

Deep Learning-Based Prostate Cancer Classification

Vision Transformers, Explainable AI

- Engineered a Swin Transformer pipeline with CLAHE preprocessing, achieving 96.4% classification accuracy and 0.90 ROC-AUC on histopathological images — outperforming CNN baselines by ~8%.
- Integrated Gradio-based explainable AI visualization to highlight diagnostic decision regions, enabling clinician trust and supporting scalable hospital deployment.

Finova AI – Financial Stability & Scheme Recommendation Platform

React, Node.js, PostgreSQL, Firebase, AI-based Risk Analysis

- Developed an AI-driven platform to detect early debt risk and map users to relevant government schemes (PMJDY, Mudra, SHG, MSME), enabling financial inclusion for underserved communities.
- Built a responsive React frontend with seamless API integration to a Node.js backend, enabling real-time financial analysis, risk scoring, and personalized guidance.

IMPRINTS — Fingerprint-Based Blood Group Identification

EfficientNet, Web App

- Built a non-invasive blood group prediction system using EfficientNet, achieving 85.84% accuracy — deployed as a real-time web app for emergency and rural healthcare settings.
- Designed for scalability in resource-constrained environments, reducing reliance on lab tests in underserved communities.

Smart Agriculture IoT System

NodeMCU, IoT, Real-Time Analytics | 1st Place, University Expo 2024

- Deployed IoT sensor network + automated irrigation system with real-time soil moisture analytics via Blynk, reducing manual oversight and optimizing water usage.
- Won 1st place at Dayananda Sagar University project expo (2024) and selected as a Global Nominee at NASA Space Apps Challenge 2024 from thousands of international submissions.

Hybrid Piezo-Solar Smart Road for Sustainable EV Charging

ESP32, IoT, Wireless Power Transfer, Renewable Energy | Academic Project

- Deployed IoT sensor network + automated irrigation system with real-time soil moisture analytics via Blynk, reducing manual oversight and optimizing water usage.
- Won 1st place at Dayananda Sagar University project expo (2024) and selected as a Global Nominee at NASA Space Apps Challenge 2024 from thousands of international submissions.

Adaptive Audio Steganography (STC + GOAS)

2nd Place, University Expo 2025

- Designed a high-fidelity steganography system using Syndrome-Trellis Codes and Hamming coding, embedding encrypted messages in audio with robust imperceptibility metrics.

PUBLICATIONS

- "Multimodal Machine Learning Approaches for Early Detection and Classification of Breast Cancer Using Imaging and Genomic Data" — Published 2025
- "Automated License Plate Recognition Using Deep Learning for Smart Mobility" — Published 2026
- "Hybrid Piezo-Solar pad with wheel embedded receiver for sustainable EV charging" — Published 2026

EDUCATION

Dayananda Sagar University

B.Tech — Computer Science & Engineering (Data Science) 2026 | **CGPA: 9.42 / 10**

Christ Academy Junior College

Class XII (PCMB) 2022 | **89.3%**

De Sales Academy

Class X (ICSE) 2020 | **92%**

ACHIEVEMENTS & AWARDS

- 1st Place — Project Expo, Smart Agriculture IoT System, Dayananda Sagar University (2024)
- Global Nominee — NASA Space Apps Challenge 2024 ('Leveraging Earth Observation Data for Informed Agricultural Decision-Making')
- 2nd Place — Project Expo, Adaptive Audio Steganography, Dayananda Sagar University (2025)
- Best SDG Innovation — 36 Hours Hackathon-CodeSangram, Alliance University (2026)